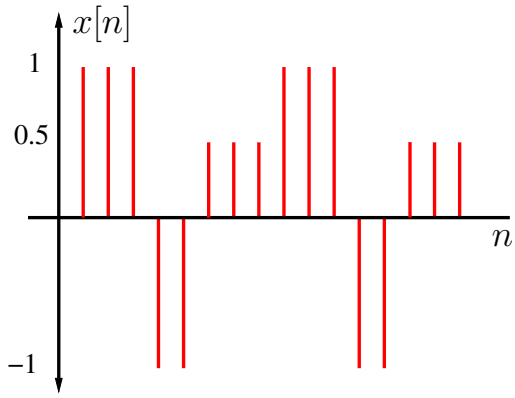
Model Answer Assignment-2 EE322M

1. Find the following

(a) Fourier series coefficients of the following periodic signal with period $N = 8$.



Solution (a): Note that starting the signal index at $n = 0$ gives us

$$
\begin{aligned}
x[0] &= \frac{1}{2}, \\
x[1] &= x[2] = x[3] = 1, \\
x[4] &= x[5] = -1, \\
x[6] &= x[7] = \frac{1}{2}.
\end{aligned}
$$

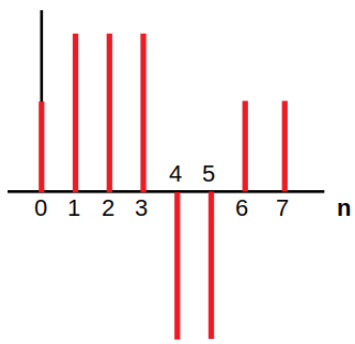


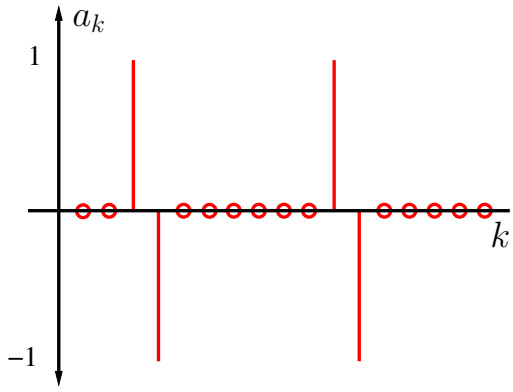
Figure 1:

Hence, using the formula to compute the spectral coefficients: $a_k = \frac{1}{8}\sum_{n=0}^{7} x[n]e^{-j\frac{2\pi}{8}kn}$, we have

$$
\begin{aligned}
a_0 &= 0.3125, \\
a_1 &= 0.3200825 + 0.2834709j, \\
a_2 &= -0.25 - 0.1875j, \\
a_3 &= 0.0549175 + 0.1584709j, \\
a_4 &= -0.0625, \\
a_5 &= 0.0549175 - 0.1584709j \\
a_6 &= -0.25 + 0.1875j, \\
a_7 &= 0.3200825 - 0.2834709j.
\end{aligned}
$$

You can use the code written for Question 11 to find the spectral coefficients. ■

(b) Discrete-time periodic signal that admits the following periodic spectral coefficients with period $N = 8$.



Solution 1(b): Note that starting the signal index at $k = 0$ gives us

$$
\begin{aligned}
&x[0] = x[1] = x[2] = 0,\\
&x[3] = 1\\
&x[4] = -1,\\
&x[5] = x[6] = x[7] = 0.
\end{aligned}
$$

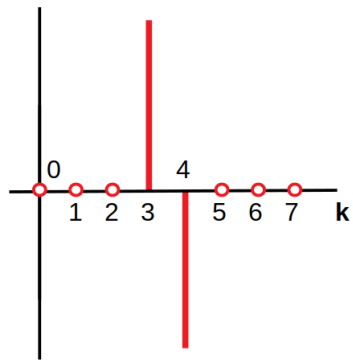


Figure 2:

Hence, using the formula to compute the spectral coefficients: $a_k = \sum_{n=0}^{7} x[n]e^{-j\frac{2\pi}{8}kn}$, we have

$$
\begin{aligned}
x[n] = \{\underset{\uparrow}{0}, 0.2928932 - 0.7071068j, -1. + 1j, 1.7071068 - 0.7071068j,\\
-2, 1.7071068 + 0.7071068j, -1 - 1j, 0.2928932 + 0.7071068j\}.
\end{aligned}
$$

■

2. The following facts about a real signal $x(t)$ with Laplace transform $X(s)$ are given:

(a) $X(s)$ has exactly two poles.

(b) $X(s)$ has no zeros in the finite s-plane.

(c) $X(s)$ has a pole at $s = -1 + j$.

(d) $e^{2t}x(t)$ is not absolutely integrable.

(e) $X(0) = 8$.

Determine $X(s)$.

Solution: From (a) and (b) we know that $X(s)$ is of the form

$$X(s) = \frac{A}{(s+a)(s+b)}. \tag{1}$$

Furthermore, it is given that one of the poles of $X(s)$ is $-1+j$. Since $x(t)$ is real, poles of $X(s)$ must occur in conjugate reciprocal pairs. Therefore, the other pole has to be $-1-j$. Thus, we must have $a = 1-j$, and $b = 1+j$. Using this in equation (1), we have

$$X(s) = \frac{A}{(s+1-j)(s+1+j)}. \tag{2}$$

From (e), we know that $X(0) = 8$. Hence evaluating $X(s)$ at $s = 0$, we have from equation (2)

$$X(0) = 8 = \frac{A}{(1-j)(1+j)} \Rightarrow A = 16.$$

Therefore, we have

$$X(s) = \frac{16}{s^2+2s+2}. \tag{3}$$

Let $\mathcal{R}$ denote the ROC of $X(s)$. From the pole locations we know that there are two possible choices of $\mathcal{R}$. $\mathcal{R}$ may either be $\mathrm{Re}(s) < -1$ or $\mathrm{Re}(s) > -1$. Statement (d) will be used to pick the correct ROC.

We know that

$$y(t) = e^{2t}x(t) \Rightarrow Y(s) = X(s-2).$$

The ROC of $Y(s)$ is $\mathcal{R}$ shifted by 2 to the right. Since it is given that $y(t)$ is not absolutely integrable, the ROC of $Y(s)$ should not include the $j\omega$-axis. This is possible only if $\mathcal{R}$ is $\mathrm{Re}(s) > -1$. ■

3. In the following figures (Figure 3 and Figure 4) the pole-zero configuration of two discrete-time systems are given.

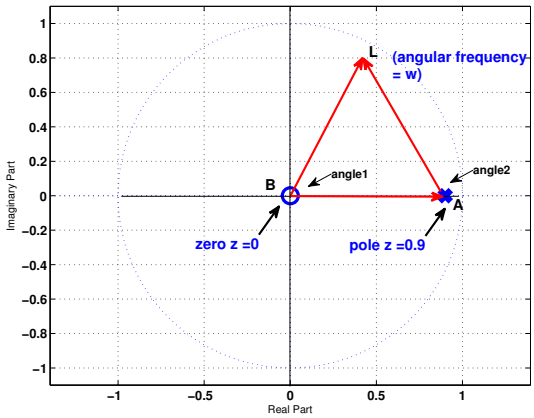


Figure 3: Pole-zero plot of System A

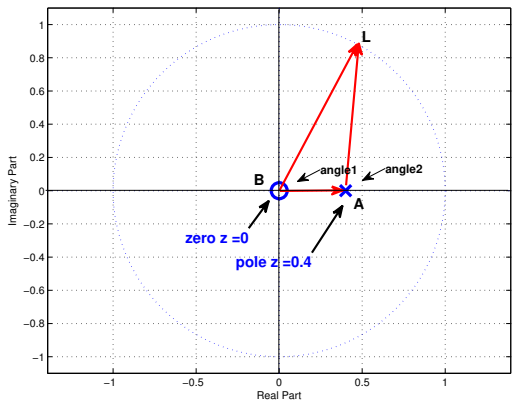


Figure 4: Pole-zero plot of System B

(a) Find the magnitude and phase response of both the systems from the graphical representation using pole vectors and zero vectors.

(b) Compute the magnitude and phase response of both the systems directly from the transfer function $H(z)$ derived from the pole-zero plot.

(c) Comment on the magnitude and phase response of both the systems. (Notice things like what happens when the poles are near imaginary axis, when is the magnitude response maximum/minimum, etc.)

Solution: (a)

System A: From the pole-zero plot: one pole at $z = 0.9$ and one zero at $z = 0$, thus the transfer function is $H(z)$

$$H(z) = \frac{z}{z - 0.9}$$

Now,

$$H(e^{j\omega}) = \frac{e^{j\omega}}{e^{j\omega} - 0.9}.$$

The magnitude response at frequency ω is given by the product of the lengths of the vectors drawn from the zeros to the point $e^{j\omega}$ divided by the lengths of vectors drawn from the poles to $e^{j\omega}$. Now, in our case, at a particular frequency ω, i.e. point L in the unit circle, the pole vector (AL) and zero vector (BL) can be computed as,

$$\begin{aligned} \mathbf{BL} &= \mathbf{BA} + \mathbf{AL} \\ \Rightarrow \mathbf{AL} &= \mathbf{BL} - \mathbf{BA} = e^{j\omega} - 0.9. \end{aligned}$$

and zero vector will be, $\mathbf{BL} = e^{j\omega}$.
Thus,

$$H(e^{j\omega}) = \frac{|\mathbf{BL}|\measuredangle\mathbf{BL}}{|\mathbf{AL}|\measuredangle\mathbf{AL}}$$

Now, at $\omega = 0$, the magnitude response

$$|H(e^{j0})| = \frac{|\mathbf{BL}|}{|\mathbf{AL}|} = \frac{1}{|e^{j0} - 0.9|} = 10$$

Similarly,

$$\left|H(e^{j\omega})\right|_{\omega=\frac{\pi}{4}} = \frac{1}{|\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} - 0.9|} = 1.4,$$

$$\left|H(e^{j\omega})\right|_{\omega=\frac{\pi}{2}} = \frac{1}{|j - 0.9|} = 0.74,$$

$$\left|H(e^{j\omega})\right|_{\omega=\frac{3\pi}{4}} = \frac{1}{|-\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} - 0.9|} = 0.57,$$

$$\left|H(e^{j\omega})\right|_{\omega=\pi} = \frac{1}{|-1 - 0.9|} = 0.52.$$

The plot for the magnitude response is given below in Figure 5.
Now we find the phase response.

System B: From the pole-zero plot: one pole at $z = 0.4$ and one zero at $z = 0$, thus the transfer function is $H(z)$

$$H(z) = \frac{z}{z - 0.4}$$

Now,

$$H(e^{j\omega}) = \frac{e^{j\omega}}{e^{j\omega} - 0.4}.$$

The magnitude response at frequency ω is given by the product of the lengths of the vectors drawn from the zeros to the point $e^{j\omega}$ divided by the lengths of vectors drawn from the poles to $e^{j\omega}$. Now, in our case, at a particular frequency ω, i.e. point L in the unit circle, the pole vector (AL) and zero vector (BL) can be computed as,

$$\begin{aligned} \mathbf{BL} &= \mathbf{BA} + \mathbf{AL} \\ \Rightarrow \mathbf{AL} &= \mathbf{BL} - \mathbf{BA} = e^{j\omega} - 0.4. \end{aligned}$$

and zero vector will be, $\mathbf{BL} = e^{j\omega}$.
Thus,

$$H(e^{j\omega}) = \frac{|\mathbf{BL}|\measuredangle\mathbf{BL}}{|\mathbf{AL}|\measuredangle\mathbf{AL}}$$

Now, at $\omega = 0$, the magnitude response

$$|H(e^{j0})| = \frac{|\mathbf{BL}|}{|\mathbf{AL}|} = \frac{1}{|e^{j0} - 0.4|} = \frac{10}{6}.$$

Similarly,

$$\left|H(e^{j\omega})\right|_{\omega=\frac{\pi}{4}} = \frac{1}{|\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} - 0.4|} = 1.3,$$

$$\left|H(e^{j\omega})\right|_{\omega=\frac{\pi}{2}} = \frac{1}{|j - 0.4|} = 0.92,$$

$$\left|H(e^{j\omega})\right|_{\omega=\frac{3\pi}{4}} = \frac{1}{|-\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} - 0.4|} = 0.76,$$

$$\left|H(e^{j\omega})\right|_{\omega=\pi} = \frac{1}{|-1 - 0.4|} = 0.71.$$

The plot for the magnitude response is given below in Figure 6.

System A: Clearly,

$$\begin{aligned}\mathbf{AL} &= e^{j\omega} - 0.9 \\ &= \cos\omega + j\sin\omega - 0.9 \\ &= (\cos\omega - 0.9) + j\sin\omega, \\ \mathbf{BL} &= e^{j\omega}.\end{aligned}$$

Therefore,

$$\begin{aligned}\measuredangle H(e^{j\omega}) &= \measuredangle\mathbf{BL} - \measuredangle\mathbf{AL} \\ &= \omega - \tan^{-1}\left(\frac{\sin\omega}{\cos\omega - 0.9}\right).\end{aligned}$$

The plot for the phase response is shown in figure below in Figure 5.

System B: Clearly,

$$\begin{aligned}\mathbf{AL} &= e^{j\omega} - 0.4 \\ &= \cos\omega + j\sin\omega - 0.4 \\ &= (\cos\omega - 0.4) + j\sin\omega, \\ \mathbf{BL} &= e^{j\omega}.\end{aligned}$$

Therefore,

$$\begin{aligned}\measuredangle H(e^{j\omega}) &= \measuredangle\mathbf{BL} - \measuredangle\mathbf{AL} \\ &= \omega - \tan^{-1}\left(\frac{\sin\omega}{\cos\omega - 0.4}\right).\end{aligned}$$

The plot for the phase response is shown in figure below in Figure 6.

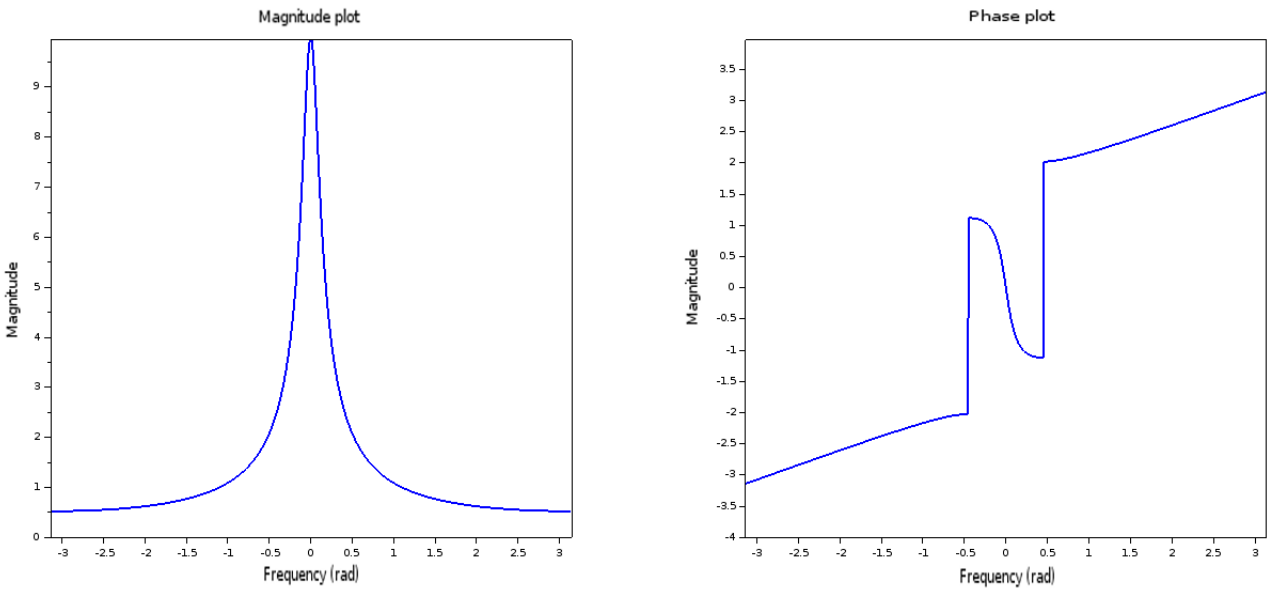


Figure 5: Magnitude and phase response of System A (Question 3a)

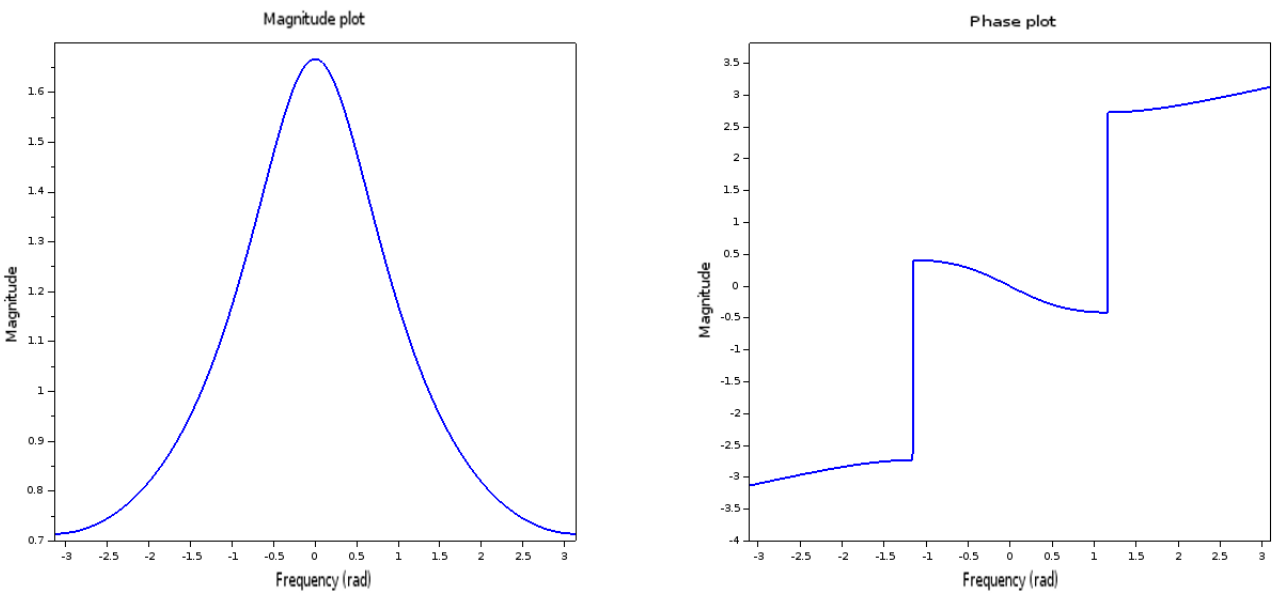


Figure 6: Magnitude and Phase response for System B (Question 3a)

Solution (b): For System A: The magnitude response is,

$$H(e^{j\omega}) = \frac{e^{j\omega}}{e^{j\omega} - 0.9} \Rightarrow |H(e^{j\omega})| = \frac{|e^{j\omega}|}{|e^{j\omega} - 0.9|} = \frac{1}{|\cos\omega + j\sin\omega - 0.9|} = \frac{1}{\sqrt{1.81 - 1.8\cos\omega}}$$

and phase response,

$$\measuredangle H(e^{j\omega}) = \omega - \tan^{-1}\left(\frac{\sin\omega}{\cos\omega - 0.9}\right).$$

For System B: The magnitude response is,

$$H(e^{j\omega}) = \frac{e^{j\omega}}{e^{j\omega} - 0.4} \Rightarrow |H(e^{j\omega})| = \frac{|e^{j\omega}|}{|e^{j\omega} - 0.4|} = \frac{1}{|\cos\omega + j\sin\omega - 0.4|} = \frac{1}{\sqrt{1.16 - 0.8\cos\omega}}.$$

and the phase response,

$$\measuredangle H(e^{j\omega}) = \omega - \tan^{-1}\left(\frac{\sin\omega}{\cos\omega - 0.4}\right).$$

Solution (c): Comments –

- From Figure 5 it is clear that both the pole-zero plots represents low pass filter.
- As the pole reaches to vicinity of the unit circle, the magnitude response decays more sharply.
- The bandwidth of the filter decreases as pole nears the unit circle.
- As the pole nears the unit circle maximum phase shift occurs near the zero frequency.

■

4. Let $h(t)$ be the impulse response of a causal and stable LTI system with a rational system function.

(a) Is the system with impulse response $\frac{d}{dt}h(t)$ guaranteed to be causal and stable?

(b) Is the system with impulse response $\int_{-\infty}^{t} h(t)d\tau$ guaranteed to be causal and unstable?

Solution: Since $h(t)$ is causal and stable, all the poles of $H(s)$ are in open left half $\mathbb{C}$ plane, where $H(s)$ is the Laplace transform of $h(t)$.
(a) The Laplace transform of $g(t) := \frac{dh(t)}{dt}$ is $G(s) := sH(s)$. Since $H(s)$ has no poles on the imaginary axis, $G(s)$ has the same poles as $H(s)$. Therefore, $G(s)$ have all poles in the open left-half $\mathbb{C}$ plane. Therefore, $g(t)$ is also guaranteed to be causal and stable.
(b) Let $r(t) := \int_{-\infty}^{t} h(t)d\tau$. Using the integration property of Laplace transform, the Laplace transform of $r(t)$ is $R(s) = \frac{H(s)}{s}$. $R(s)$ does not have a pole at $s = 0$ only if $H(s)$ has a zero at $s = 0$. Therefore, we cannot guarantee that $r(t)$ is always causal & stable. ■

5. Find the impulse response of an LTI system, having the following properties.

(a) The system is causal.

(b) The system is rational, and has only two poles, at $s = -2$ and $s = 4$.

(c) If input $x(t) = 1$, then output $y(t) = 0$.

(d) $h(0^+) = 4$, i.e., the value of the impulse response at $t = 0^+$ is 4.

Solution: From fact (a) and fact (b), we assume the transfer function to be of the following form:

$$H(s) := \frac{p(s)}{(s+2)(s-4)} \tag{4}$$

Recall that corresponding to an input $x(t) = e^{st}$ to an LTI system, the output is $y(t) = H(s)e^{st}$. Here we have $x(t) = 1 = e^{0.t}$. Therefore, $s = 0$ here. The output is corresponding to $x(t) = 1$ is $H(0)e^{0.t} = H(0)$. Thus, we must have according to fact (c) in the question

$$H(0) = 0 \Rightarrow \frac{p(0)}{2 \times (-4)} = 0 \Rightarrow p(0) = 0.$$

This means that $p(s)$ must have zero as a root. Without loss of generality, we assume $p(s) = sq(s)$, where $q(s)$ is a polynomial in s. Next we use fact (d). By initial value theorem:

$$h(0^+) = \lim_{s\to\infty} sH(s) = \lim_{s\to\infty} \frac{s^2 q(s)}{(s+2)(s-4)} = 4. \tag{5}$$

Observe that if the numerator has higher degree than the denominator, the limit will diverge. Hence, $q(s)$ needs to be independent of s. Let $q(s) = K$. Then, we have from equation (5), we have

$$\lim_{s\to\infty} \frac{s^2 K}{(s+2)(s-4)} = 4 \Rightarrow K = 4. \tag{6}$$

Thus, we have $H(s) = \frac{4s}{(s+2)(s-4)}$. ■

6. A method to build a DC power supply is to take an AC signal and full-wave rectify it. That is, we put the AC signal $x(t)$ through a system that produces $y(t) = |x(t)|$ as its output.

(a) Sketch the input and output waveforms if $x(t) = \sin t$. What are the fundamental periods of the input and output?

(b) If $x(t) = \sin t$, determine the coefficients of the Fourier series for the output $y(t)$.

(c) What is the amplitude of the DC component of the input signal? What is the amplitude of the DC component of the output signal?

Solution: I am providing the solution for the more general case, i.e., assuming $x(t) = A\sin(\omega_1 t)$, with $A > 0$.
(a) The full wave rectified output is shown in Figure 7 with respect to input sine wave.

The fundamental period of $x(t)$ is $T_1 = 2\pi/\omega_1$. From Figure 7 it is clear that the period of the rectified sinusoid $y(t) = A|\sin(\omega_1 t)|$ is $T = T_1/2 = \pi/\omega_1$.

(b) Define ω_0 to be the fundamental angular frequency of $y(t)$, therefore $\omega_0 := 2\pi/T$. Observe that ω_1 is related to ω_0 by the following relation:

$$\omega_1 = 2\pi/T_1 = \pi/T = \omega_0/2.$$

Thus we can write, $y(t) = A|\sin(\frac{\omega_0 t}{2})|$ The Fourier series coefficients are

$$c_k = \frac{1}{T}\int_0^T y(t)e^{-jk\omega_0 t}dt = \frac{1}{T}\int_0^T A|\sin(\omega_0 t/2)|e^{-jk\omega_0 t}dt$$

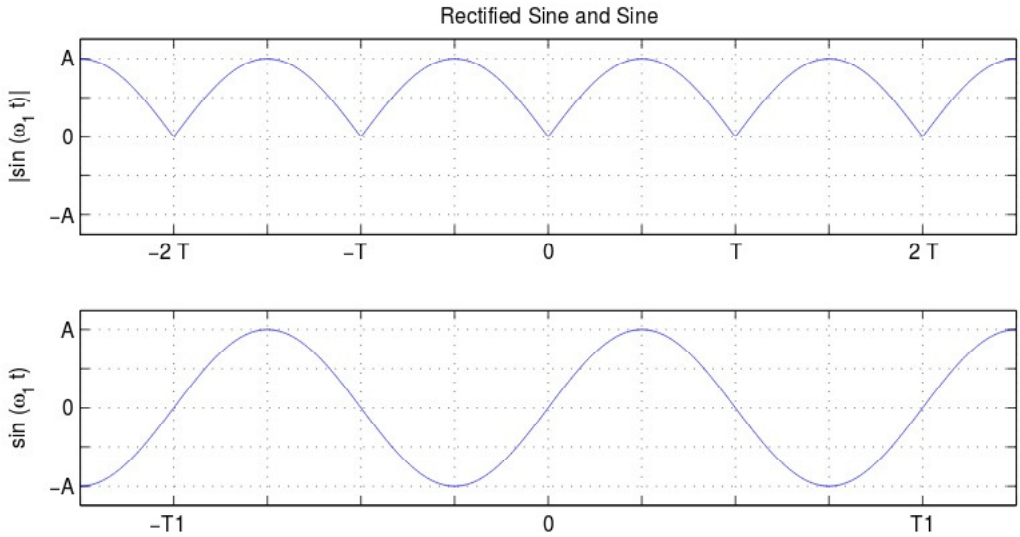


Figure 7: *Half wave rectified and input sine wave.*

From 0 to T, $|\sin(\omega_0 t/2)| = \sin(\omega_0 t/2)$. Also, replace ω_0 by π/T:

$$\begin{aligned} c_k &= \frac{1}{T}\int_0^T A\sin(\omega_0 t/2)e^{-jk\pi t/T}dt = \frac{A}{T}\int_0^T \left(\frac{e^{j\pi t/T} - e^{-j\pi t/T}}{j2}\right)e^{-jk\pi t/T}dt \\ &= -\frac{A}{2T}\left[\frac{e^{j\pi(1-2k)/T}}{\pi(1-2k)/T} + \frac{e^{-j\pi(1+2k)/T}}{\pi(1+2k)/T} - \frac{1}{\pi(1-2k)/T} - \frac{1}{\pi(1+2k)/T}\right] \end{aligned}$$

But, $e^{j\pi(1-2k)} = e^{j\pi}e^{-j2\pi k}$. $e^{j\pi} = -1$ and $e^{-j2\pi k} = 1$. Thus,

$$\begin{aligned} c_k &= -\frac{A}{2T}\left[\frac{-1}{\pi(1-2k)/T} + \frac{-1}{\pi(1+2k)/T} - \frac{1}{\pi(1-2k)/T} - \frac{1}{\pi(1+2k)/T}\right] \\ &= \frac{2A}{\pi}\left[\frac{1}{(1-4k^2)}\right] \end{aligned}$$

(c) For the DC component we need to check for $k = 0$, since this corresponds to zero frequency. Therefore, the DC component of the output is $c_0 = \frac{2A}{\pi}$. (For the question given, just put $A = 1$) The DC component of the input is zero. ■

7. Consider a continuous-time LTI system whose frequency response is

$$H(j\omega) = \begin{cases} 1, & |\omega| \geqslant 100, \\ 0, & \text{otherwise.} \end{cases}$$

When the input to this system is a signal $x(t)$ with fundamental period $T = \frac{\pi}{8}$ and Fourier series coefficients a_k, it is found that the output $y(t)$ is identical to $x(t)$. For what values of k is it guaranteed that $a_k = 0$?

Solution: Let b_k be the Fourier series coefficients of the output $y(t)$. Therefore, $b_k = H(jk\omega_0)a_k$, where $\omega_0 = \frac{2\pi}{T} = 16$ and a_k are the Fourier series coefficients of $x(t)$.

If $y(t)$ is identical to $x(t)$, then $b_k = a_k$ for all k. Note that since $H(j\omega) = 0$ for $|\omega| < 100$, we must have $H(jk\omega_0) = H(j16k) = 0$ for $|k| < 7$. Therefore, $a_k = 0$ for $|k| < 7$. ■

8. Suppose that a continuous-time periodic signal is the input to an LTI system. The signal has a Fourier series representation

$$x(t) = \sum_{k=-\infty}^{\infty} \alpha^{|k|} e^{jk\frac{\pi}{4}t},$$

where α is a real number between 0 and 1, and the frequency response of the system is

$$H(jw) = \begin{cases} 1, & |\omega| \leqslant W, \\ 0, & |\omega| > W. \end{cases}$$

How large must W be in order for the output of the system to have at least 90% of the average energy per period of $x(t)$?

Solution: The average energy per period by Parseval's theorem is

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2 = \sum_{k=-\infty}^{\infty} \alpha^{2|k|} = 2 \times \sum_{k=0}^{\infty} \alpha^{2|k|} - 1 = \frac{2}{1-\alpha^2} - 1 = \frac{1+\alpha^2}{1-\alpha^2}.$$

We want N such that

$$\begin{aligned} \sum_{k=-N}^{N} |a_k|^2 \geqslant 0.9 \times \frac{1+\alpha^2}{1-\alpha^2} &\Rightarrow \sum_{k=-N}^{N} \alpha^{2|k|} \geqslant 0.9 \times \frac{1+\alpha^2}{1-\alpha^2} \\ &\Rightarrow 2 \times \sum_{k=0}^{N} \alpha^{2|k|} - 1 \geqslant 0.9 \times \frac{1+\alpha^2}{1-\alpha^2} \\ &\Rightarrow \frac{1 - 2\alpha^{2(N+1)} + \alpha^2}{1-\alpha^2} \geqslant 0.9 \times \frac{1+\alpha^2}{1-\alpha^2} \\ &\Rightarrow N \geqslant \frac{\log[0.05(1+\alpha^2)]}{2\log\alpha} - 1 \end{aligned}$$

Hence, the window width must be

$$W \geqslant \frac{\pi N}{4} \Rightarrow W \geqslant \frac{\pi \log[0.05(1+\alpha^2)]}{8\log\alpha} - \frac{\pi}{4}.$$

■

9. Consider a moving average system with an impulse response given by:

$$h[n] = h[0]\delta[n] + h[1]\delta[n-1] + h[2]\delta[n-2],$$

where $h[0] = h[1] = h[2] = \frac{1}{3}$.

(a) Find the frequency response (both magnitude and phase) of the system.

(b) How do the magnitude and phase plots change if a unit time-delay is introduced to the system?

Solution: (a) The transfer function of the moving average system is

$$H(z) = h[0] + h[1]z^{-1} + h[2]z^{-2} = \frac{1}{3}\left(1 + z^{-1} + z^{-2}\right).$$

In order to find the frequency response, we evaluate the transfer function at $z = e^{j\omega}$. Thus, we have

$$H(e^{j\omega}) = \frac{1}{3}\left(1 + e^{-j\omega} + e^{-2j\omega}\right) = \frac{e^{-j\omega}}{3}\left(e^{j\omega} + 1 + e^{-j\omega}\right) = \frac{e^{-j\omega}}{3}\left(1 + 2\cos\omega\right)$$

Therefore,

$$|H(e^{j\omega})| = \frac{1}{3}|1+2\cos\omega| \qquad \angle H(e^{j\omega}) = \begin{cases} -\omega & \cos\omega \geqslant -\frac{1}{2} \\ \pi - \omega & \cos\omega < -\frac{1}{2}. \end{cases}$$

Solution: (b) On introduction of unit-delay the transfer function of the system becomes:

$$\widehat{H}(j\omega) = z^{-1}H(z) = \frac{1}{3}\left(z^{-1} + z^{-2} + z^{-3}\right).$$

Therefore, the frequency response is

$$\widehat{H}(e^{j\omega}) = e^{-j\omega}H(e^{j\omega}) = \frac{e^{-j2\omega}}{3}\left(1+2\cos\omega\right).$$

Therefore,

$$|H(e^{j\omega})| = \frac{1}{3}|1+2\cos\omega| \qquad \angle H(e^{j\omega}) = \begin{cases} -2\omega & \cos\omega \geqslant -\frac{1}{2} \\ \pi - 2\omega & \cos\omega < -\frac{1}{2}. \end{cases}$$

■

10. Let the following periodic signal

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t-3k) + \delta(t-1-3k) - \delta(t-2-3k)$$

be the input to an LTI system with transfer function

$$H(s) = e^{s/4} - e^{-s/4}.$$

Find the spectral coefficients of the resulting output signal $y(t)$.
Solution: The spectral coefficient of $x(t)$ is

$$a_k = \frac{1}{T}\int_T x(t)e^{-j\omega_0 kt}dt.$$

The signals looks like that in Figure 8. From the figure, it is clear that $T = 3$ and $\omega_0 = \frac{2\pi}{3}$.

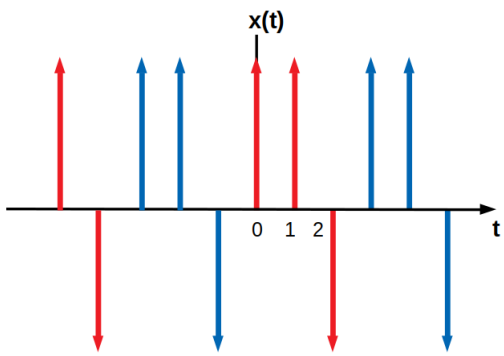


Figure 8:

Therefore,

$$a_k = \frac{1}{3}\int_0^3 [\delta(t) + \delta(t-1) - \delta(t-2)]e^{-j\frac{2\pi}{3}kt}dt = \frac{1}{3}\left[1 + e^{-j\frac{2\pi}{3}k} + e^{-j\frac{2\pi}{3}2k}\right]$$
$$= \frac{1}{3}\left[1 + e^{-j\frac{2\pi}{3}k} + e^{j\frac{2\pi}{3}k}\right] = \frac{1}{3}\left[1 + 2\cos\left(\frac{2\pi}{3}\right)\right].$$

Note that on using $s = j\omega$, we have

$$H(j\omega) = e^{j\omega/4} - e^{-j\omega/4} = 2j \sin\left(\frac{\omega}{4}\right).$$

Thus, the spectral coefficients of the output $y(t)$ is

$$b_k = H(j\omega_0 k)a_k = 2j \sin\left(\frac{\frac{2\pi}{3}k}{4}\right) \times \frac{1}{3}\left[1 + 2\cos\left(\frac{2\pi}{3}\right)\right] = \frac{2j}{3}\sin\left(\frac{\pi k}{6}\right)\left[1 + 2\cos\left(\frac{2\pi}{3}\right)\right].$$

■